Secretin	Acid Fat	S cells of the duodenum, jejunum, and ileum	Stimulates Pepsin secretion Pancreatic Bicarbonate secretion Biliary bicarbonate secretion Growth of exocrine pancreas Inhibits Gastric acid secretion
Gastric inhibitory peptide	Protein Fat Carbohydrate	K cells of the duodenum and jejunum	Stimulates Insulin release Inhibits Gastric acid secretion
Motilin	Fat Acid Nerve	M cells of the duodenum and jejunum	Stimulates Gastric motility Intestinal motility

There are five major hormones that help in digestion. They are

1. Gastrin is secreted by the G cells of the gastric pits and it stimulates the gastric glands to secrete pepsinogen and HCl. The stimulation for gastrin secretion is food in the stomach and it is inhibited by low pH. Gastrin also stimulates the growth of the mucosa.
2. The presence of acidic chyme in the duodenum stimulates the secretion of a hormone called Secretin, which in turn stimulates the pancreas to secrete a bicarbonate rich fluid. It also stimulates the secretion of bile in the liver.
3. Cholecystokinin (CCK) is another hormone in the duodenum that stimulates the release of digestive enzymes from the pancreas and also it is involved in the emptying of bile in the gallbladder. This hormone is secreted in response to the abundance of fat in chyme.
4. Enterogastrone and Gastric inhibitory peptide (GIP) released from the duodenum decreases the churning action of stomach and slows the emptying of the stomach. It also induces insulin secretion.
5. Motilin is a duodenal hormone that regulates gastrointestinal motility and stimulates the production of pepsin.
6. Others: Hepatocrinin stimulates bile formation with lesser bile salts. Enterocrinin is involved in the stimulation of the secretion of enzymes by the intestinal mucosa. Chymodenin stimulates the secretion of chymotrypsin from pancreas.

Appetite-Regulating Hormones

Ghrelin, a hormone which is released by the stomach targets the pituitary gland, signaling to eat. The action of Ghrelin is countered by PYY released by the small intestine signals that you have eaten enough and helps to suppress your appetite.

XII U2 Digestion.indd 51 4/8/2020 3:05:31 PM